Sources: Stein-Shakarchi, Lang, Ahlfors, Davenport x2, and Fulton-Harris **Date: 03/01/2021**

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1 Real Analysis

1.1 Fourier Analysis

Question : Define the Fourier Transform on $L^1(\mathbb{R})$.

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i\xi x}\, dx.$$

Question : Where does the Fourier transform land?
It lands in $L^\infty(\mathbb{R})$. To see this note that

$$|\hat{f}(\xi)| \leq \int_{\mathbb{R}} |f(x)|\, dx = \|f\|_{L^1} < \infty.$$

Question : Show that the Fourier Transform goes to zero as $|\xi| \to \infty$.
This is the Riemann-Lebesgue lemma. Let $f \in L^1$. Then there is a step function g s.t. $\|f - g\|_{L^1} < \epsilon$, since step functions are dense in simple functions which are dense in L^1. Since g is a step function, there is an N such that for $|\xi| > N$,

$$\left| \int_{\mathbb{R}} g e^{ix\xi}\, dx \right| < \epsilon.$$

Indeed, suffices to show this for a characteristic function $g(x) = \chi_{[a,b]}(x)$ and use linearity. Anyways, so now we can just write

$$\int_{\mathbb{R}} f e^{i\xi x}\, dx = \int_{\mathbb{R}} (f(x) - g(x))e^{i\xi x}\, dx + \int_{\mathbb{R}} g(x)e^{i\xi x}\, dx.$$

Taking absolute values on both sides we obtain

$$\left| \int_{\mathbb{R}} f e^{i\xi x}\, dx \right| <= \int_{\mathbb{R}} |f(x) - g(x)|\, dx + \left| \int_{\mathbb{R}} g(x)e^{ix\xi}\, dx \right| < 2\epsilon,$$

for $|\xi| > N$. This shows that $\lim_{|\xi|\to\infty} \int f(x)e^{i\xi x}\, dx = 0$.
Question : Show that the Fourier Transform is continuous
Let $\xi_n \to \xi$. Then define $f_n(x) = f(x)e^{i\xi_n x}$. Then note that $|f_n(x)| \leq |f(x)| \in L^1(\mathbb{R})$. By the Dominated convergence theorem

$$\lim_{n\to\infty} \int_{\mathbb{R}} |f_n - f| = 0,$$

, so $\hat{f}(\xi)$ is continuous.
Question : We have shown that the Fourier transform is an operator from $L^1(\mathbb{R})$ to $C_0(\mathbb{R})$ with the infinity norm on C_0. Show it is not surjective.
Ok so first let $g_n = \frac{\sin(2\pi x)\sin(2\pi n x)}{\pi^2 x^2}$ and note that $f_n * f_1$ is it's Fourier transform, where

$f_n = 1_{[-n,n]}$. Indeed the Fourier transform of a convolution is the product of the Fourier transforms, so we can take Fourier transforms of f_n and f_1. We obtain

$$\hat{f}_n(y) = \int_{-n}^{n} e^{2\pi ixy}\, dx = \frac{\sin(2\pi ny)}{\pi y},$$

and we obtain something similar for $\hat{f}_1$. Now suppose the transform was surjective. Then it is actually bijective, and by the open mapping theorem it has a bounded inverse. But since $\|g_n\|_{L^1} \to \infty$ as $n \to \infty$, such an inverse can't be bounded.

Question : Why does the L^1 norm of g_n go to infinity?

Oh ok, well we write

$$\begin{aligned}\|g_n\|_{L^1} &= \int_{\mathbb{R}} \frac{|\sin(2\pi x)\sin(2\pi nx)|}{\pi^2 x^2}\, dx = \int_{\mathbb{R}} \frac{|\sin(x)\sin(x/n)|}{\pi^2 x^2}\frac{2n}{\pi}\, dx \\ &= \frac{4n}{\pi}\int_0^{\infty} \frac{|\sin(x)\sin(x/n)|}{\pi^2 x^2}\, dx.\end{aligned}$$

Then note that for $t \in [0,1]$ $\sin(t) \geq t - t^3/6 \geq \frac{5}{6}t$ so $\sin(x/n) \geq \frac{5}{6}\frac{y}{n}$ on $[0,n]$. So we obtain

$$\|g_n\|_{L^1} \gg \int_0^n \frac{|\sin(x)|}{x}\, dx \to \infty \text{ as } n \to \infty.$$

Question : Write down $f(x) = \sum_{n=N}^{2N} e^{2\pi in^2x}$. What is the L^2 norm of f on the circle?

We have

$$\begin{aligned}\|f\|_{L^2}^2 &= \int_0^1 |f(x)|^2\, dx = \int_0^1 f(x)f(\bar{x})\, dx \\ &= \int_0^1 \sum_{n,m=N}^{2N} e^{2\pi i(n^2-m^2)x}\, dx = N+1.\end{aligned}$$

So $\|f\|_{L^2} = \sqrt{N+1}$.

Question : How about the L^4 norm?

We know that $\|f\|_{L^4}^4 = \left\|f^2\right\|_{L^2}^2$. Expanding we have

$$\left\|f^2\right\|_{L^2}^2 = \sum_{\nu_1^2-\nu_3^2=\nu_2^2-\nu_4^2} 1 \ll 2N^2 + N^2N^{\epsilon},$$

where the first term comes from the diagonal terms and the second term is found by fixing $\nu_2^2 - \nu_4^2$ and bounding the number of solutions by the number of factors of $\nu_2^2 - \nu_4^2$, using the divisor function and the bound $\tau_k(n) \ll n^{\epsilon}$ for any k. So we have $\|f\|_{L^4} \ll N^{1/2+\epsilon}$.

Question : How can we use this to get a lower bound for the L^1 norm?

We use Holder's inequality. We have

$$\int_0^1 |f|^2 \leq \left(\int_0^1 |f|\right)^{2/3}\left(\int_0^1 |f|^4\right)^{1/3} \ll \|f\|_{L^1}^{2/3}\left(N^2 + N^2N^{\epsilon}\right)^{1/3},$$

so we have

$$\|f\|_{L^1} \gg \left(\frac{N^{1/2}}{N^{2/3+\epsilon}}\right)^{3/2} = N^{1/2-\epsilon}.$$

Question : What is Plancherel's Theorem?
It says that the Fourier transform on $L^2(\mathbb{R})$ is an isometry. In other words,

$$\left\|\hat{f}\right\|_{L^2} = \|f\|_{L^2}.$$

Question : Let me give you f in $L^2(\mathbb{R})$, C^1 with L^2 norm equal to 1, and its Fourier transform supported on, say, a ball of radius n. Can you give a bound on the L^2 norm of f'?
Since the Fourier transform is an isometry on L^2, we know that

$$\|f'\|_{L^2} = \left\|\hat{f'}\right\|_{L^2} = \left\|2\pi i\xi\hat{f}\right\|_{L^2} \leq 2\pi n\left\|\hat{f}\right\|_{L^2} = 2\pi n\|f\|_{L^2} = 2\pi n.$$

I don't even know if this bound is correct.

1.2 Measure Theory

Question : What is Lebesgue measure?
Let μ^* denote the Lebesgue outer measure and let $E \subseteq \mathbb{R}^n$. Then E is Lebesgue measurable if and only if

$$\mu^*(A) = \mu^*(E \cap A) + \mu^*(E^c \cap A),$$

for every $A \subseteq \mathbb{R}^n$. This is the Caratheodory criterion.
Question : What is a measurable function?
A function is measurable if it pulls back measurable sets to measurable sets.
Question : What is a Borel set?
Take the closure of open sets under complements, countable unions, and countable intersections.
Question : Are there more measurable sets than Borel sets?
Yes. The quick answer is to construct a homeomorphism from $[0,1]$ to $[0,2]$ that will map a subset of the Cantor set of measure zero to a set of measure one. The set of measure one must contain a non-measurable set N. The preimage of this set N will be Lesbegue measurable, but not Borel.
Now for the whole proof. Let $f : [0,1] \to [0,2]$ with $f(x) = c(x) + x$ where $c(x)$ is the Cantor function which takes value $1/2$ on the middle third, $1/4$ on the middle third on $(0,1/3)$ and $3/4$ on the middle third on $(2/3,1)$, and so on... Note that $f' = 1$ so f is strictly increasing. Also f is continuous. Moreoever, f is injective, and it's surjective since $f(0) = 0$ and $f(1) = 2$ it takes every value in-between by the intermediate value theorem. Not hard to see f^{-1} is continuous, so we have a homeomorphism. Now note that f maps intervals removed by the Cantor set to intervals of the same length. So

$$\mu(f([0,1]) \setminus C) = \mu([0,1] \setminus C) = 1.$$

Moreover

$$[0,2] = f(C) \cup f([0,1] \setminus C) \implies \mu(f(C)) = 1.$$

So $f(C)$ contains a non-measurable subset N. We see that $f^{-1}(N)$ is measurable and has measure zero since it is a subset of the Cantor set. Since homeomorphisms map Borel sets to Borel sets, $f^{-1}(N)$ is not Borel since N is not even measurable, hence not

Borel.

Question : Can you show every measurable set $A \subset \mathbb{R}$ with positive measure contains a non-measurable set?

Without loss of generality, we may assume $A \subset (0,1)$. Now define an equivalence relation by $x \sim r$ iff $x - r \in \mathbb{Q}$. The elements of A are hence partitioned by this relation. Let N be the set obtained by picking exactly one representative from each equivalence class. We claim N is not measurable. Suppose N was measurable. Note that $A \subset \cup_{r\in(-1,1)\cap\mathbb{Q}} N + r$. Indeed, if $x \in A$, then either $x \in N$ or x differs from an element of N by a rational number. It is also a disjoint union since if $N + r \cap N + s$ were nonempty, then we'd have $n_1 + r = n_2 + s$ giving $n_1 - n_2 \in \mathbb{Q}$, a contradiction. Let $\{r_i\}_{i=1}^{\infty}$ be an enumeration of the rationals. Then we have

$$\mu(A) \leq \sum_{i=1}^{\infty} \mu(N + r_i) = \sum_{i=1}^{\infty} \mu(N) \leq 3. \tag{1}$$

This is a contradiction, so N is non-measurable.

Question : What is the Arzela-Ascoli theorem? Can you prove it?

It says that if I have a sequence of continuous functions on a closed and bounded interval that are uniformly bounded and uniformly equicontinuous, then there is a subsequence that converges uniformly. The idea of the proof is a diagonalization argument. Let $\{x_i\}$ be an enumeration of the rational numbers in the interval I. Then $\{f_n(x_i)\}$ has a convergent subsequence for each fixed i, by applying Bolzano-Weierstrass since the $\{f_n\}$ are uniformly bounded. Then construct a sequence $\{f\}$ using the diagonal terms which will converge at every rational. For this sequence we have for a given $\epsilon > 0$ and x_k,

$$|f_n(x_k) - f_m(x_k)| < \epsilon/3,$$

for $n, m \geq N$. Now by the uniform equicontinuity, we know that for this same ϵ, for every $x \in I$ there is a δ such that $|f(s) - f(t)| < \epsilon/3$ whenever $s, t \in U_\delta(x)$. Now note that the the $U_\delta(x)$ form an open cover of I and so by compactness we can take a finite subcover. Moreover, there is a K so that each $U_j(x)$ contains a rational x_k with $1 \leq k \leq K$. So for any $t \in I$, there is a neighborhood U_j and a k so that t and x_k live in the same ball. Then we have

$$|f_n(t) - f_m(t)| \leq |f_n(t) - f_n(x_k)| + |f_n(x_k) - f_m(x_k)| + |f_m(x_k) - f_m(t)| < \epsilon,$$

for $n, m \geq N = \max\{N(\epsilon, x_1), \ldots, N(\epsilon, x_K)\}$.

Question : What is the Radon-Nikodym theorem?

It says that if μ is a σ-finite measure and ν is a σ-finite signed measure, then if $\nu \ll \mu$ there is a μ-integrable f such that

$$\nu(E) = \int_E f(x)\, d\mu(x).$$

Question : What is $(L^p)^*$?

It is L^q where $1/p + 1/q = 1$. The idea is to take a linear functional $\ell \in (L^p)^*$ and define

$$\nu(E) = \ell(\chi_E).$$

Then it is simple to see that $|\nu(E)| \leq \|\ell\| \, (\mu(E))^{1/p}$. This implies that ν is absolutely continuous with respect to μ. By Radon-Nikodym, there is an integrable g such that

$$\nu(E) = \int \chi_E g \, d\mu,$$

which gives $\ell(\chi_E) = \int \chi_E g \, d\mu$. By the linearity of ℓ this extends to simple functions, and since simple functions are dense in L^p, we obtain $\ell(f) = \int fg \, d\mu$. A bit of computation shows that $\|\ell\| = \|g\|_q$.

Question : Prove completeness of L^p.

Pick a subsequence so that $\left\|f_{n_i} - f_{n_{i+1}}\right\|_p < 2^- i$. Then let

$$g_n(x) = \sum_{i=1}^{n} |f_{n_{i+1}}(x) - f_{n_i}(x)|$$
$$g(x) = \sum_{i=1}^{\infty} |f_{n_{i+1}}(x) - f_{n_i}(x)|.$$

Note that by Minkowski's inequality $\|g_n\|_p \leq 1$. Then by Fatou's lemma

$$\int |g|^p \leq \liminf \int |g_n|^p \leq 1,$$

so $\|g\|_p \leq 1$. Then the function

$$f_{n_1}(x) + \sum_{i=1}^{\infty} |f_{n_{i+1}}(x) - f_{n_i}(x)|,$$

converges pointwise a.e. to a function $f \in L^p$. Indeed,

$$\lim_{i\to\infty} f_{n_i} = \lim_{i\to\infty} \left(f_{n_1} + \sum_{k=1}^{i-1} (f_{n_{k+1}}(x) - f_{n_k}(x)) \right) = f.$$

Question : Give a nowhere differentiable continuous function

Consider the Weierstrass function $f(x) = \sum_{n=1}^{\infty} a^n \cos(b^n \pi x)$ where $|a| < 1$ and $ab > 1 + 3\pi/2$. The a^n terms give the uniform continuity and the b^n provides the oscillation which makes getting a derivative impossible.

1.3 Functional Analysis

Question : State the Bounded Inverse Theorem

If X and Y are Banach spaces and $T\colon X \to Y$ is a continuous bijective linear transformation, then $T^{-1}\colon Y \to X$ is also continuous and in particular bounded.

Question : Give a counterexample to show this doesn't hold if the spaces aren't complete.

Let X be the space of sequences $x\colon \mathbb{N} \to \mathbb{R}$ wich only finitely many nonzero terms equipped with supremeum norm. Then

$$\begin{aligned} T\colon X &\to X \\ (x_n) &\mapsto (x_1, x_2/2, x_3/3, \dots) \end{aligned}$$

is bounded, linear, but T^{-1} is unbounded. To see X is not complete observe that $x^{(n)} = (1, 1/2, \dots, 1/n, 0, \dots, 0)$ converges as $n \to \infty$ to $x^{\infty} = (1, 1/2, \dots, 1/n, \dots)$ which does not live in X since it has infinitely many nonzero terms.

Question : What is the Baire category theorem?

Any complete metric space with nonempty interior (or any subset of one) is not the countable union of nowhere dense sets.

Question : Can you use it to prove the uniform boundedness principle?

Sure. So the uniform boundedness principle says that if $\mathcal{F}$ is a collection of continuous linear operator from a Banach space X to a normed space Y and $\sup_{T\in\mathcal{F}} \|T(x)\|_Y < \infty$ for every $x \in X$, then $\sup_{T\in\mathcal{F}} \|T\| < \infty$. To prove it let $X_n := \{x \in X : \sup_{T\in\mathcal{F}} \|T(x)\|_Y \leq n\}$. Then each X_n is closed and by assumption we have $X = \cup_{n\geq 1} X_n$. By the Baire Category theorem, these can't be all nowhere dense, so there exists m such that X_m has non-empty interior. So there is an $x_0 \in X_m$ and an $\epsilon > 0$ such that the closed ball $B_\epsilon(x) \subseteq X_m$. So then for any unit vector $\|u\| \leq 1$ and $T \in \mathcal{F}$, we have

$$\begin{aligned} \|T(u)\|_Y &= \frac{1}{\epsilon} \|T(x_0 + u\epsilon) - T(x_0)\|_Y \\ &\leq \frac{1}{\epsilon} (\|T(x_0 + u\epsilon)\|_Y + \|T(x_0)\|_Y) \leq 2m\epsilon^{-1} < \infty. \end{aligned}$$

Taking the supremum over u in the unit ball and $T \in \mathcal{F}$ gives the result.

Question : What does the Hahn-Banach theorem say?

Let V be a vector space, let W be a subspace, and let $\ell \in W^*$ such that

$$|\ell(x)| \leq p(x) \ \forall x \in W,$$

where $p : V \to \mathbb{R}$ is a nonnegative sublinear function. Then there exists $\varphi \in V^*$ such that

$$\varphi|_W = \ell \text{ and } |\varphi(x)| \leq p(x) \ \forall x \in V.$$

Question : How are separability and duality related?

If X is a Banach space and X^* is separable, then X is separable. The idea is to choose a dense set $\{\ell_n\} \in X^*$ and a sequence $\{x_n\} \in X$ with $\|x_n\| = 1$ such that $|\ell_n(x_n)| \geq \|\ell_n\|/2$. Then you can consider the set Y of all finite $\mathbb{Q}$-linear combinations of x_i. This is countable and we claim it is dense. If not, then there is a $y \in X \setminus Y$ and by Hahn-Banach there is a linear functional ℓ such that $\ell(x) = 0$ for all x, but on y it is nonzero. So then let $\{\ell_{n_k}\}$ be a subsequence converging to ℓ. Then we have

$$\|\ell - \ell_{n_k}\| \geq |(\ell - \ell_{n_k})(x_{n_k}| = |\ell_{n_k}(x_{n_k})| \geq \|\ell_{n_k}\|/2.$$

However this implies that $\|\ell_{n_k}\| \to 0$ as $k \to \infty$ which implies $\ell = 0$, a contradiction. Thus Y is a countable dense subset of X, so X is separable.

2 Algebra

2.1 Group Theory

Question : What is a semidirect product?

Given groups H and K and a homomorphism $\varphi\colon K \to \mathrm{Aut}(H)$ we can define G to be the set of ordered pairs (h, k) with multiplication

$$(h_1, k_1) \cdot (h_2, k_2) = (h_1\varphi(k_1)(h_2), k_1k_2).$$

The group G is denoted $H \rtimes_\varphi K$.

Question : What does this tell us about classifying groups of order n

Well, there is a theorem that says that if G is a group with subgroups H and K with H normal in G and $H \cap K = 1$, then $HK = H \rtimes_\varphi K$ where $\varphi\colon K \to \mathrm{Aut}(H)$ maps k to the automorphism of left conjugation by k on H, (which lands in H since H is normal).

Question : Ok so how do you use this to classify groups of order pq, with $p < q$?

Since $q > p$, there is a unique Sylow-q-subgroup of order q that is normal in G. Let P be a Sylow-p-subgroup and we have that $G \cong Q \rtimes P$, for some $\varphi : P \to \mathrm{Aut}(Q)$. Note that $\mathrm{Aut}(Q)$ is cyclic of order $q-1$. If $p \nmid q-1$, then there is only the trivial homomorphism and we obtain the cyclic group of order pq. If $p \mid q-1$, then $\varphi(x) = y^k$ for some $0 \leq k \leq p-1$, where x generates P and y generates the unique subgroup of order p in $\mathrm{Aut}(Q)$. When $k = 0$ we get the trivial homomorphism and hence the cyclic group of order pq. When $k \neq 0$ all the resulting nontrivial homomorphisms give isomorphic groups G_k which are the nonabelian semidirect products.

Question : What are Sylow's theorems?

Let G be a group of order $p^\alpha m$. Then Sylow's theorems state that there exists a Sylow-p-subgroup of order p^α, that any two Sylow-p-subgroups are conjugate, and that if n_p denotes the number of Sylow-p-subgroups, $n_p \equiv 1 \bmod p$ and $n_p \mid m$.

Question : Can you classify groups of order 30?

Sure. So let G be a group of order 30 which contains a group H of order 15. Since $3 \nmid (5-1)$, H is cyclic and H is normal because it has index 2. By Sylow's theorems there is a subgroup K of order 2, so $G = HK$ and $H \cap K = 1$ so $G \cong H \rtimes_\varphi K$ for $\varphi : K \to \mathrm{Aut}(H)$ a homomorphism. We have that $\mathrm{Aut}(H) = \mathrm{Aut}(\mathbb{Z}/15\mathbb{Z}) = (\mathbb{Z}/15\mathbb{Z})^* = (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$. Since $K = (\mathbb{Z}/2\mathbb{Z})$, $\varphi(1)$ must generate a subgroup of order 2 in $\mathrm{Aut}(H)$. There are three subgroups of order 2, which act on $H := \langle a \rangle \times \langle b \rangle$ by sending a to a^{-1}, b to b^{-1} or both. Let $K = \langle k \rangle$. Consider the automorphism where the action of k is given by $k \cdot a = a$ and $k \cdot b = b^{-1}$. Then k centralizes the element a, so we can see G as $G \cong H \rtimes_\varphi K = \langle a \rangle \times \langle b, k \rangle$ where $b^3 = 1$ and $k^2 = 1$. So $G \cong \mathbb{Z}/5\mathbb{Z} \times D_6$. For the flipped automorphism we obtain $G \cong \mathbb{Z}/3\mathbb{Z} \times D_{10}$. Finally the automorphism sending both a and b to their inverse corresponds to the group D_{30} and the trivial automorphism corresponds to $\mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$.

Question : What can you say about the center of a group of order p^n?

Well the center of sucha group is nontrivial. By the class equation, we have that

$$|G| = |Z(G)| + \sum_i [G\colon C_G(x_i)],$$

where $Z(G)$ is the center and $C_G(x_i)$ is the conjugacy class of a represenative x_i. Since p divides the LHS it should also divide the RHS. Since each $[G\colon C_G(x_i)]$ is divisible by p, we must have $p \mid |Z(G)|$, so the center is nontrivial.

Question : Can you show a group of order p^2 is Abelian?

Let $\rho\colon G \to \mathrm{GL}(V)$ be a representation of G. Suppose $V \cong \oplus_i V_i$. Then we have that $|G| = \sum_i (\dim V_i)^2$. Since the trivial representation always exists, it must be that $\dim V_i = 1$ for all i and that there are p^2 such spaces. But then every representation of G is one-dimensional, so G is abelian.

Question : Can you compute the character table for A_4?

First we compute the commutator subgroup $[A_4, A_4] = \{e, (12)(34), (13)(24), (14)(23)\}$ which is normal and $A_4/[A_4, A_4] \cong (\mathbb{Z}/3\mathbb{Z})$. Then the conjugacy classes come from the conjugacy classes of S_4, where odd cycles split. For the representations, the characters of $(\mathbb{Z}/3\mathbb{Z})$ are well understood. The last character must be of dimension 3 by a counting argument, or we can see that it is the standard representation of S_4 restricted to A_4. We could also use orthogonality relations to compute the last row. In any case, the character table we obtain is

A_4	e	(123)	(132)	(12)(34)
χ_0	1	1	1	1
χ_1	1	ω	ω^2	1
χ_2	1	ω^2	ω	1
Std	3	0	0	-1

Question : How do you show the commutator is normal?
Commutator subgroup is generated by $[a,b] = a^{-1}b^{-1}ab$. Conjugating this element we have

$$ga^{-1}b^{-1}abg^{-1} = (ga^{-1}g^{-1})(gb^{-1}g^{-1})(gag^{-1})(gbg^{-1}) = (a')^{-1}(b')^{-1}(a')(b') \in [G,G].$$

2.2 Galois Theory

Question : What is the fundamental theorem of Galois theory?
If G is the Galois group of an extension L/K, then for every intermediate field $K \subset E \subset L$, there is a subgroup H of G that are in 1-to-1 correspondence. Given a subgroup H, you pick E to be L^H. Given a intermediate field E, the subgroup is $\mathrm{Aut}(E/K)$.
Question : What is the Galois group of $x^5 - 2$ over $\mathbb{Q}$? The splitting field is $\mathbb{Q}(2^{1/5}, \omega)$ where ω is a fifth root of unity. Note that it has order 20 since the subextension $\mathbb{Q}(2^{1/5})$ is degree 5 over $\mathbb{Q}$ and the extension by ω is degree 4. The Galois group G has order 20 and so by Sylow's theorems the Sylow-5-subgroup is normal. Consider the explicit automorphisms given by

$$\sigma\colon \begin{cases} 2^{1/5} & \mapsto \omega 2^{1/5} \\ \omega & \mapsto \omega \end{cases} \text{ and } \tau\colon \begin{cases} 2^{1/5} & \mapsto 2^{1/5} \\ \omega & \mapsto \omega^2 \end{cases}.$$

Then the subgroup generated by σ is cyclic of order 5, and the subgroup generated by τ is cyclic of order 4. Now we know G is of the form $(\mathbb{Z}/5\mathbb{Z}) \rtimes_\varphi (\mathbb{Z}/4\mathbb{Z})$ where $\varphi : (\mathbb{Z}/4\mathbb{Z}) \to \mathrm{Aut}((\mathbb{Z}/5\mathbb{Z})$ maps τ to the conjugation automorphism sending σ to $\tau^{-1}\sigma\tau$. We can check that this automorphism is nontrivial and so the semidirect product is not direct. In particular G is not abelian.
Question : Can you give an Abelian Galois group of order 20?
Take $\mathrm{Gal}(\mathbb{Q}(\zeta_{44})/\mathbb{Q})$ where ζ_{44} is a 44th root of unity.
Question : What is the Galois group of $x^3 - 2$ over $\mathbb{Q}$
The Galois group G has order 6 so it is either S_3 or $(\mathbb{Z}/6\mathbb{Z})$. To show it is S_3 it suffices to show G is not Abelian. Consider the automorphisms

$$\sigma\colon \begin{cases} 2^{1/3} & \mapsto \omega 2^{1/3} \\ \omega & \mapsto \omega \end{cases} \text{ and } \tau\colon \begin{cases} 2^{1/3} & \mapsto 2^{1/3} \\ \omega & \mapsto \omega^2 \end{cases}.$$

Then $\sigma\tau(2^{1/3}) = \omega 2^{1/3}$ while $\tau\sigma(2^{1/3}) = \omega^2 2^{1/3}$.
Question : When is the Galois group of a number field a subset of A_n?
Let L be the splitting field of the min polynomial, and α_i its roots. Then $\mathrm{Gal}(L/K) \subset A_n$

when $\prod_{i<j}(\alpha_i - \alpha_j) \in K$. Indeed, note that the product is fixed by $\sigma \in \mathrm{Gal}(L/K)$ if and only if the number of transpositions is even (because each transposition would introduce a minus sign in the product). But this is true if and only if $\sigma \in A_n$, when viewed as an element of S_n. This also tells us that the discriminant must be a perfect square, since it is the square of the product above.

Question : What is the Hilbert class field of $\mathbb{Q}(\sqrt{-5})$? Prove it.

The field $L = \mathbb{Q}(\sqrt{-5}, i)$ is unramified over $K = \mathbb{Q}(\sqrt{-5})$. Indeed, consider the quadratic subfields $\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{-5}), \mathbb{Q}(i)$. The only primes that could possibly ramify in L are 2 and 5 by considering the discriminant. Now a priori, the ramification degree could be 4, but then such a prime would have to ramify in both $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(i)$, but no prime ramifies in both. So the ramification degree must be 2. But both 2 and 5 ramify in K/Q. So by multiplicativity of the ramification degrees, the ramification in L/K must be 1.

2.3 Rings

2.4 Representations

Question : What is a representation of a group?

A representation of a group G on a complex vector space V is a homomorphism

$$\rho\colon G \to GL(V).$$

Question : What is Mashke's theorem? Prove it.

It says that every representation of a finite group G on a finite dimensional complex vector space is a direct sum of irreducible representations. Suppose W is a G-invariant subspace of V. Then given any Hermitian inner product on V, we can construct a G-invariant Hermitian inner product by averaging. In particular, we can obtain

$$\langle v, w \rangle := \frac{1}{|G|} \sum_{g \in G} \{gv, gw\},$$

where $\{,\}$ is the original inner product. Then it's easy to see that $W^\perp$ is also G-invariant and we obtain $V \cong W \oplus W^\perp$. The result follows by induction.

Question : What is Schur's Lemma? Let $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(V')$ be two irreducible representations. Let $T\colon V' \to V$ is a linear transformation such that the following diagram commutes:

$$\begin{tikzcd} V' \arrow[r, "T"] \arrow[d, "\rho'(g)"] & V \arrow[d, "\rho(g)"] \\ V' \arrow[r, "T"] & V \end{tikzcd}$$

Then T is either bijective or zero. Moreover, if $V \cong V'$ then T is a scalar map.

Question : What can you say about the dimensions of the irreducible representations of a finite group G?

If $\rho_1, \dots, \rho_r$ are isomorphism classes of irreducible representations of G and $\chi_1, \dots, \chi_r$ are the characters , then the dimension d_i of ρ_i (or χ_i) divides $|G|$ and moreover

$$|G| = d_1^2 + \cdots + d_r^2.$$

To prove it, consider the action of the regular representation.

Question : Show that if G is a finite abelian group, then all the irreducible

representations are one-dimensional
Let $\rho : G \to \mathrm{GL}(V)$ be an irreducible representation and let $g \in G$. Since G is Abelian, $\rho(g)\rho(h) = \rho(h)\rho(g)$. Now if we think of $\rho(g)$ as a linear transformation T, then we have $\rho(h)(T(v)) = T(\rho(h)(v))$, which means the diagram from Schur's Lemma commutes. So by Schur's lemma, $\rho(g) = T$ is scalar. So each $g \in G$ acts as a scalar, so every line in V is a G-invariant subspace. Since V is irreducible, V itself must be one-dimensional.

2.5 Modules

Question : State the Fundamental Theorem for Finitely Generated Modules over a PID.
Let R be a PID and let M be a finitely generated R-module. Then M is isomorphic to the direct sum of finitely many cyclic modules, namely

$$M \cong R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m),$$

for nonzero elements a_i of R which are not units and satisfy $a_1 \mid a_2 \mid \cdots \mid a_m$.
Question : How do you get rational canonical form?
Let V be an n-dimensional vector space over the field F. Since F is a field, $F[x]$ is a PID. Let $T\colon V \to V$ be a linear transformation. Then V has a natural structure as an $F[x]$ module by sending $(x, v) \mapsto Tv$. By the fundamental theorem, we have that

$$V \cong F[x]/(p_1) \oplus \cdots \oplus F[x]/(p_r).$$

We can assume the p_i are monic. Suppose $p_1 = \sum_{i=0}^{k} a_i x^i$. Then the action of T on the basis $\{1, x, \ldots, x^{k-1}\}$ gives $\{x, x^2, \ldots, -a_0 - a_1 x - a_2 x^2 - \ldots a_{k-1}x^{k-1}\}$. With respect to the basis, the matrix for multiplication by x is therefore

$$\begin{pmatrix} 0 & 0 & \ldots & \ldots & \ldots & -a_0 \\ 1 & 0 & \ldots & \ldots & \ldots & -a_1 \\ 0 & 1 & \ldots & \ldots & \ldots & -a_2 \\ 0 & 0 & \ddots & & & \vdots \\ \vdots & \vdots & & \ddots & & \vdots \\ 0 & 0 & \ldots & \ldots & 1 & -a_{k-1} \end{pmatrix}.$$

If A is the $n \times n$ matrix for the linear map T, then the above is a block matrix. The matrix for A will decompose into r block matrices and its form is called rational canonical form.
Question : How would you classify the conjugacy classes of $\mathrm{GL}_2(\mathbb{F}_p)$?
Well a matrix A determines an $\mathbb{F}_p[x]$-module structure on $M = \mathbb{F}_p^2$. Two matrices are conjugate if and only if they give rise to the same module structure. If M is not cyclic, then M is of the form $\mathbb{F}_p[x]/(x-a) \oplus \mathbb{F}_p[x]/(x-a)$ in which case the resulting rational canonical form of A is

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}.$$

If M is not cyclic then it is of the form $\mathbb{F}_p[x]/(f(x))$ where $f(x)$ is the characteristic polynomial for A. If $f(x) = x^2 + ax + b$ is reducible, then we obtain the rational canonical form matrix

$$\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}.$$

If $f(x) = (x-a)(x-b)$ we have the matrix

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

Finally if f has a repeated root $f(x) = (x-a)^2$, we obtain

$$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}.$$

By a simple counting argument this shows there are $q^2 - 1$ conjugacy classes.
Question : What does the number of conjugacy classes tell you?
It is the number of irreducible representations of $\mathrm{GL}_2(\mathbb{F}_p)$ over the complex numbers. (Think of a character table for example)
Question : What is Jordan canonical form?
It's like rational canonical form in the case where we are working over a algebraically closed field. For example, we might consider a $\mathbb{C}$-vector space $V \cong \mathbb{C}^n$ and given a linear transformation T we can give V a $\mathbb{C}[x]$-module structure. Since $\mathbb{C}$ is algebraically closed, the summands in the direct sum decomposition of V will be of the form $\mathbb{F}_p[x]/(p_i)^k$ where p_i is a linear polynomial $(x-a)$. Then picking the basis $\{(x-a)^{k-1}, \ldots, (x-a), 1\}$ gives matrices of the form

$$\begin{pmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & a \end{pmatrix},$$

in the case $k = 3$ for example.

3 Complex Analysis

3.1 Harmonic Function Theory

Question : Prove that every harmonic function on a simply connected domain is the real part of a holomorphic function
Let u be our harmonic function and let $g(z) = \partial_x u - i\partial_y u$. Then g satisfies the Cauchy-Riemann equations and hence is holomorphic. So there exists a primitive for g in Ω, our simply connected domain, given by

$$f(z) = u(z_0) + \int_\gamma g(z)\, dz,$$

where γ is a path from z_0 to z. If U is the real part of f, then we have

$$\partial_x U - i\partial_y U = f'(z) = g(z) = \partial_x u - i\partial_y u.$$

So U and u agree up to a constant since their gradients are the same. However since $U(z_0) = u(z_0)$, $U = u$, hence u is the real part of the holomorphic function f.
Question : Why do you need the domain to be simply connected? What can go wrong?
We can consider the function $u = \log|z|$ which is harmonic in $\mathbb{C}\backslash 0$, but is not the real part of any holomorphic function on $\mathbb{C}\backslash 0$. Indeed the function would be $f(z) = \log|z| + i\arg(z)$ which is not well defined on $\mathbb{C} \setminus 0$.

Question : Prove the mean value property for harmonic functions.
Well since a harmonic function u is the real part of a holomorphic function f, the Cauchy integral formula tells us that

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{i\theta})\,d\theta,$$

and then taking the real parts of both sides gives the result.
Question : Show that a bounded harmonic function on $\mathbb{C}$ is constant.
Let u be the harmonic function. It is the real part of a holomorphic function f. So $|e^{f(z)}| = e^u \leq e^M$ since it is bounded. So $e^{f(z)}$ is bounded and entire, so it is constant. So f must be constant, and hence u.
Question : How would you do this on $\mathbb{R}^n$?
You use the mean value property. Let $x, y \in \mathbb{R}^n$ and let $r = R + d(x, y)$. Thenm $B_R(x) \subset B_r(y)$. Indeed $z \in B_R(x)$, so $d(z, x) < R$ but then

$$d(z, y) \leq d(z, x) + d(x, y) < R + (r - R) = r.$$

Using the mean value property, we have

$$u(x) = \frac{1}{\mu(B_R(x))}\int_{B_R(x)} u(x)\,dx \leq \frac{1}{\mu(B_R(x))}\int_B r(y)u(x)\,dx,$$

so

$$u(x) \leq \frac{\mu(B_r(y))}{\mu(B_R(x))}u(y) \leq \frac{(R + d(x,y))^n}{R^n}u(y).$$

Taking $R \to \infty$, we obtain that $u(x) \leq u(y)$. Then by symmetry, we can do the same thing to conclude $u(y) \leq u(x)$. Since x, y were arbitrary u is constant.
Question : Suppose I have a subharmonic function in the plane bounded above by 3, what can you say about it?
It's constant. Let $u(z)$ be the subharmonic function. Let $M = \max_{z\in\mathcal{D}} |u(z)|$. By the maximum principle, $M = \max_{z\in\partial\mathcal{D}} |u(z)|$. Let $v(z) = u(z) - \epsilon \log|z|$. There are three important properties about $v(z)$. One, it is subharmonic away from zero because $\log|z|$ is harmonic away from zero. Secondly, it agrees with u on the unit circle and third it tends to $-\infty$ as $|z| \to \infty$. By the maximum principle for subharmonic functions, we have

$$M = \max_{z\in\mathcal{D}} v(z) = \max_{z\in\partial\mathcal{D}} v(z) = \max_{z\in\partial\mathcal{D}} u(z).$$

In fact, $|v(z)| \leq M$ for $z \geq 1$ as well, since $v(z)$ tends to $-\infty$. Letting ϵ go to zero, we can make the same conclusion for $u(z)$. Hence $|u(z)| \leq M$ for all z, yet attains its maximum on the interior. So it must be constant everywhere.
Question : Draw a semicircle. Suppose I have a holomorphic function f bounded above by 2 on the circular part and 1 on the diameter. Give a bound for f in the interior.
It suffices to find a harmonic function u such that $|u(z)| = \log 2$ on the circular part and zero on the diameter. Then $\log|f(z)|$ is subharmonic and bounded above by $u(z)$ on the boundary of the semicircle. By the maximum principle, $\log|f(z)| \leq |u(z)|$ on the interior (this is applying maximum principle to the difference of the functions). Now to find the function $u(z)$, let $v(z)$ be the angle subtended at z by the diameter. Then $v(z) = \pi/2$ on the circular part and π on the diameter. So define $u(z) = \frac{\log 2}{\pi/2}(\pi - v(z))$.

Question : Prove the maximum principle for harmonic functions.
Let Ω be a region. We'd like to show that if u is a non-constant harmonic function, then $\max_{z\in\Omega}|u(z)| = \max_{z\in\partial\Omega}|u(z)|$, i.e. the maximum of $|u(z)|$ is attained on the boundary. Suppose u attains it's max at a point x_0 in the interior. Then there exists $\delta > 0$ such that $B_\delta(x_0)$ is contained in the interior. By the mean value property, $u(x_0) = \frac{1}{2\pi}\int_0^{2\pi} u(x_0 + \delta e^{i\theta})\, d\theta$. However we know that $u(x_0 + \delta e^{i\theta})$ is strictly less than the maximum M. So we have

$$M = |u(x_0)| < \frac{1}{2\pi}\int_0^{2\pi} M = M,$$

so $u(z)$ must be constant.

3.2 Regular Complex

Question : What is Liouville's theorem?
It says that any bounded entire function is constant.
Question : Can you prove it?
Sure, so for any z, let γ be the circle of radius R centered around z. Suppose $|f(z)| \le M$. Then by the Cauchy integral formula, we have

$$f'(z) = \int_\gamma \frac{f(\zeta)}{(\zeta - z)^2}\, d\zeta,$$

and a quick estimatte gives

$$\begin{aligned} |f'(z)| &= \left|\int_\gamma \frac{f(\zeta)}{(\zeta - z)^2}\, d\zeta\right| = \left|\int_0^{2\pi} \frac{f(\zeta)}{(\zeta - z)^2} iRe^{it}\, dt\right| \\ &\le \int_0^{2\pi} \frac{M}{R}\, dt = O(1/R). \end{aligned}$$

Since f is entire, we take $R \to \infty$ which shows the derivative vanishes, so f is constant.
Question : Prove the fundamental theorem of algebra
Let $P(z)$ be a non-constant polynomial of degree n. Assume it has no zeros. Then $\frac{1}{P(z)}$ is a bounded holomorphic function. Indeed if $P(z) = a_n z^n + \cdots + a_0$ then

$$\frac{P(z)}{z^n} = a_n + (a_{n-1}/z + \cdots + a_0/z^n)\,.$$

Taking $|z| \to \infty$, shows that there exists R such that $|P(z)| \ge |a_n|/2|z|^n$ whenever $|z| > R$. By Liouville's theorem, $1/P(z)$ is constant.
Question : How do you evaluate the sum of the reciprocals of the squares?
The product formula for sine tells us that

$$\frac{\sin \pi z}{\pi} = z\prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right).$$

Expanding both sides and taking the coefficient of z^3, we see that $\frac{(\pi z)^3}{3!\pi} = z^3\sum_{n=1}^{\infty} 1/n^2$, which gives $\pi^2/6$.
Question : How do you get the product formula for sine?

Let $G(z) = \frac{\sin \pi z}{\pi}$ and $P(z) = z\prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right)$. Then note that $\frac{G'(z)}{G(z)} = \frac{P'(z)}{P(z)}$ by the cotangent formula. So then

$$\left(\frac{P(z)}{G(z)}\right)' = \frac{P(z)}{G(z)}\left[\frac{P'(z)}{P(z)} - \frac{G'(z)}{G(z)}\right] = 0.$$

So $P(z) = cG(z)$ for some c. But looking back at the identity this would give

$$\frac{\sin \pi z}{\pi z} = c\prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right),$$

and taking $\lim |z| \to 0$ on both sides shows $c = 1$. So $P(z) = G(z)$.

Question : Give an example of an essential singularity

Consider the function $f(z) = \exp(1/z)$. It has a singularity at $z = 0$. It is not a removable singularity because $e^{1/z}$ is not bounded as $z \to 0$. It is not a pole either because $z^n e^{1/z}$ still blows up for any n.

Question : What does the Phragmen-Lindelof theorem say? Can you prove it?

The version I know says that if $S = \{z : -\pi/4 < \arg z < \pi/4\}$ then if F is a holomorphic function such that $|F(z)| \le 1$ on the boundary of S and $|F(z)| = O\left(e^{c|z|}\right)$, then F must actually be bounded by 1 inside the sector. Indeed, the growth condition is necessary, for example e^{z^2} is bounded by 1 on the boundary, but is unbounded on the real line which is inside the sector. To prove it we construct an auxiliary function $F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}$. We note that $z^{3/2} = r^{3/2}e^{i3\theta/2}$ so if $|\theta| < \pi/4$, $3\theta/2$ still has positive cosine. So we obtain that $|F_\epsilon(z)| \le 1$ on the boundary of S and moreover, $|F_\epsilon(z)| \to 0$ as $|z| \to \infty$. Because of this decay condition, we claim that $|F_\epsilon(z)| \le 1$ inside S. The same conclusion for $|F(z)|$ will follow by taking $\epsilon \to 0$. Indeed, let $M = \sup_{z\in S}|F_\epsilon(z)|$ and because $|F_\epsilon(z)| \to 0$, we know that M is in a bounded subdomain of S. Hence we can apply the maximum modulus principle and conclude that M is attained on the boundary. But then we know that $M \le 1$ if it's on the boundary, and hence $|F_\epsilon(z)| \le 1$.

Question : What is Jensen's formula?

It says that if f is holomorphic in a disk of radius R with zeros $z_1, \ldots, z_N$ in the disk then

$$\log|f(0)| = \sum_k \log\left(|z_k/R|\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta.$$

Note that if fg satisfies the theorem so do f and g. So note we can write $f(z) = (z - z_1)\ldots(z - z_N)g(z)$ where g doesn't vanish in the disk. So it suffices to prove the theorem for functions of the form $z - w$ and functions g. First for functions g, all we need to prove is that

$$\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log\left|g(Re^{i\theta})\right|\, d\theta.$$

We can write $g = e^{h(z)}$ and then note that $|g(z)| = e^{\operatorname{Re} h(z)}$ so $\log|g(z)| = \operatorname{Re}(h(z))$ which is harmonic and thus has the mean value property. Now we deal with functions of the form $z - w$. We want to prove

$$\log|w| = \log(|w|/R) + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta.$$

It suffices to show that

$$\frac{1}{2\pi}\int_0^{2\pi} \log|e^{i\theta} - w/R|\, d\theta = 0.$$

To show this is to show that the same integral of $\log|1 - ae^{i\theta}|$ when $|a| < 1$ is zero. To see this ocnsider the function $F(z) = 1 - az$ which doesn't vanish on the disk, so there is a holomorphic function G s.t. $F(z) = e^{G(z)}$ and we use the same trick with mean value property of a harmonic function after taking logs and real parts to get the result.

Question : Can you give a bound for the number of zeros in a disk of radius r using this?

We can rewrite Jensen's formula as

$$\int_0^R \backslash(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta - \log|f(0)|$$

3.3 Mappings

Question : What is Montel's theorem? Can you prove it?

Montel's theorem says that if I have a family of holomorphic functions that is uniformly bounded on compact sets, then the family is equicontinuous and normal, meaning every sequence has a subsequence that converges uniformly on compact sets. Compare with the Arzela-Ascoli theorem which requires the family be unfiromly bounded AND equicontinuous to deduce it's normal. So we will just show that the family is equicontinuous given that it's uniformly bounded. Pick r sufficiently small so that $D_{3r}(z) \subset \Omega$ for all $z \in K$. Let γ be the circle of radius $2r$ around a point w and let $|z - w| < r$. Applying the Cauchy Integral Formula gives

$$\begin{aligned}
|f(z) - f(w)| &= \frac{1}{2\pi i}\left[\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta - \int_\gamma \frac{f(\zeta)}{\zeta - w}\, d\zeta\right] \\
&= \frac{1}{2\pi i}\int_\gamma f(\zeta)\left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w}\right) d\zeta \\
&= \frac{1}{|2\pi i|}\int_\gamma |f(\zeta)|\frac{|z - w|}{|(\zeta - z)||(\zeta - w)|}\, d\zeta \\
&\leq \frac{1}{2\pi}\int_\gamma B\frac{|z - w|}{r^2}\, d\zeta = \frac{B}{r}|z - w|,
\end{aligned}$$

so $|f(z) - f(w)| < C|z - w|$ for some constant C so it is equicontinuous. The rest of the proof is the same argument as in the proof of Arzela-Ascoli.

Question : Show that any entire injective function f from $\mathbb{C}$ to $\mathbb{C}$ takes the form $f(z) = az + b$

We look at the behavior of $f(z)$ near infinity. If $f(z)$ has a removable singularity at ∞, then f is bounded in a neighborhood of infinity, say $\{z : |z| > r\}$ for some $r > 0$. But then f is also bounded on $\{z : |z| \leq r\}$ since it is compact. So f is bounded and hence by Liouville's theorem it is constant. So f cannot have a removable singularity at ∞. Now consider f having an essential singularity at ∞. Then by the big picard theorem, f attains every value of $\mathbb{C}$ infinitely often with at most one exception. So f cannot be injective. Therefore, f has a pole at ∞, which means f is a polynomial. But the degree can't be two or higher for then there would be two values that hit zero, which is a contradiction by injectivity. So it must be a linear polynomial of the form $az + b$.

Question : Suppose $\text{Re}(f)^2 \le \text{Im}(f)^2$, for an entire function f. What can you say about f?
It is constant. Consider $e^{f(z)^2}$. This is an entire function. If we let $f = u + iv$ and expand, we see that

$$|e^{f(z)^2}| = e^{u^2 - v^2} \le 1,$$

so it is constant. This is not quite right.
Question : Prove the Casorati-Weierstrass theorem. First, what does it say?
It says that around an essential singularity z_0 of a function f, the image is dense. More specifically, if f is holomorphic in the punctured disk $D_r(z_0) - \{z_0\}$ then the image of $D_r(z_0) - \{z_0\}$ is dense in $\mathbb{C}$. Suppose it isn't dense. Then there is $w \in \mathbb{C}$ and $\delta > 0$ such that $|f(z) - w| > \delta$ for all $z \in D_r(z_0) - \{z_0\}$. Let $g(z) = \frac{1}{f(z)-w}$, which is holomorphic in the punctured disk and bounded by $1/\delta$. We examine the behavior of g near the essential singularity. If $g(z_0) \ne 0$, then $f(z) - w$ is holomorphic at z_0. If $g(z_0) = 0$, then $f(z)$ has a pole at z_0. In either case, f can't have an essential singularity at z_0, a contradiction.
Question : What does the Schwarz Lemma say?
Let $f\colon \mathcal{D} \to \mathbb{C}$ be a holomorphic map such that $f(0) = 0$ and $|f(z)| \le 1$ on $\mathcal{D}$. Then $|f(z)| \le |z|$ for all $z \in \mathcal{D}$ and $|f'(0)| \le 1$. Also if $|f(z_0)| = |z_0|$ for some z_0 or $|f'(0)| = 1$, then f is a rotation. To prove this, let

$$g(z) = \begin{cases} f(z)/z & z \ne 0 \\ f'(0) & z = 0 \end{cases}.$$

This is holomorphic on the disk. Consider the circle $|z| = r$. Then $|g(z)| = |f(z)|/|z| \le 1/r$. By the maximum principle, $|g(z)| \le 1/r$ for $z < r$ as well. Taking $r \to 1-$ we see that $|g(z)| \le 1$, which proves the first claim. Continuing, note that $f(0) = 0$ so $f(z)/z = \frac{f(z)-f(0)}{z} \le 1$ in absolute value by the above analysis. If $|f'(0)| = 1$, then g attains its max at zero and is constant. Moreover, the constant must be $|c| = 1$.
Question : Okay talk about the Riemann mapping theorem
It says that any simply connected domain that isn't all of $\mathbb{C}$ is conformally equivalent to the unit disk. In particular, for $z_0 \in \Omega$, there is a unique conformal map $F : \Omega \to \mathcal{D}$ such that $F(z_0) = 0$ and $F'(z_0) > 0$. To prove uniqueness, if there was another map G, composing with inverse gives an automorphism of the disk, which is of the form cz and the derivative condition implies $c = 1$. First we simplify the domain Ω. Our claim is that Ω is conformally equivalent to some open $\Omega' \subset \mathcal{D}$ containing 0. Let $\alpha \notin \Omega$, and note $z - \alpha$ never vanishes on Ω. By simply connectedness, we can define a single valued branch of log with function $h(z) = \log_\Omega(z - \alpha)$ which is injective. Note that for $w \in \Omega$, $h(z)$ stays away from $h(w) + 2\pi i$. If it didn't, we would have $\{z_n\} \subset \Omega$ such that $h(z_n) \to h(w) + 2\pi i$. Exponentiating would give

$$z_n - \alpha = e^{h(z_n)} \to e^{h(w)+2\pi i} = e^{h(w)} = w - \alpha,$$

which implies $z_n \to w$, so then we'd have

$$h(w) \xleftarrow{n\to\infty} h(z_n) \xrightarrow{n\to\infty} h(w) + 2\pi i.$$

So we can define

$$H(z) = \frac{1}{h(z) - (h(w) + 2\pi i)},$$

which is holomorphic, injective and bounded on Ω. That means up to rescaling and translating if necessary, $H(\Omega) = \Omega'$, an open subset of $\mathcal{D}$ containing 0.
The next step is a construction of a function f. Now we can assume the domain is a subset of $\mathcal{D}$ containing zezro. Let

$$\mathbb{F} = \{f\colon \Omega \to \mathcal{D}, f(0) = 0,\ \text{1-1 and holo }\}.$$

This set is nonempty because it contains the identity, and it is normal by Montel's theorem since $|f| < 1$ implies $\mathbb{F}$ is uniformly bounded. Let $s = \sup_{f\in\mathbb{F}} |f'(0)|$ and note that $s \geq 1$ (because the identity is in $\mathbb{F}$). We choose $f_n \in \mathbb{F}$ such that $|f_n'(0)| \to s$. Since $\mathbb{F}$ is normal, there is a subsequence converging on compact sets in Ω to $f\colon \Omega \to \bar{\mathcal{D}}$. Since f is non constant, it is $1-1$ and $|f| < 1$ on Ω by the maximum principle. Lastly $f(0) = 0$ since $f_n(0) = 0$. So $f \in \mathbb{F}$ with $|f'(0)| = s$.
The last step is to show this f is onto. We proceed by contradiction. If not, then there is $\alpha \in \mathcal{D} \setminus f(\Omega)$ and the goal is to tweak f to get some $\hat{f} \in \mathbb{F}$ with $|\hat{f}'(0)| > s$. Let ψ_α be an automorphism of the disk taking $\alpha \to 0$. Then let $U = (\psi_\alpha \circ f)(\Omega) \subset \mathcal{D}$. Note U is simply connected and $0 \notin U$ because $\alpha \notin f(\Omega)$. Now we can again define a log and square root on this domain, letting $g(w) = e^{\frac{1}{2}\log w}$. Let $F = \psi_{g(\alpha)} \circ g \circ \psi_\alpha \circ f$. This maps zero to zero to α to $g(\alpha)$ to 0. In other words, $F(0) = 0$. F is holomorphic and injective and so $F \in \mathbb{F}$. Now notice that

$$f = \psi_\alpha^{-1} \circ h \circ \psi_{g(\alpha)}^{-1} \circ F,$$

where h maps $w \to w^2$ (it is the inverse of g). Call the first three functions in the composition Φ and notice that $\Phi\colon \mathcal{D} \to \mathcal{D}$ and $\Phi(0) = 0$. By the Schwarz Lemma, $|\Phi'(0)| < 1$. Here the inequality is strict since if the derivative were exactly one, Φ would be a rotation, but Φ is not injective (because of the h factor). Then we see that

$$|F'(0)| = \frac{|f'(0)|}{|\Phi'(0)|} > |f'(0)| = s,$$

which is the contradiction we wanted.
Question : This is Koebe's proof. What was Riemann's idea for proving the mapping theorem?
Suppose f is biholomorphism from a simply connected domain Ω to the disk such that $f(z_0) = 0$. Then we can write $f(z) = (z - z_0)e^{g(z)}$ where $g(z)$ is holomorphic in Ω. The function f should map the boundary to the boundary so we have

$$1 = |z - z_0| e^{\operatorname{Re} g(z)} \text{ for all } z \in \partial\Omega.$$

Let $u(z) := \operatorname{Re} g(z)$. Then u is harmonic and we have the boundary condition $u(z) = -\log|z - z_0|$ for $z \in \partial\Omega$. So if you can solve the Diirchlet problem for Ω, you can find g and construct its harmonic conjugate to reconstruct f.
Question : How can you solve the Dirichlet problem without appealing to Riemann mapping?
You can use Perron's method. You assume the region Ω is bounded and you have a real-valued function $f(\zeta)$ on the boundary. Then you associate a harmonic function u to this f which will solve the Dirichlet problem for f. You consider a family of subharmonic functions $\mathcal{B}(f)$ such that $\limsup_{z\to\zeta} v(z) \leq f(\zeta)$ for all $\zeta \in \partial\Omega$. Then taking u to be the sup over all v in this family will solve the Dirichlet problem with boundary conditions

f.
Question : What about in the disk?
In the disk, there is an explicit solution to the Dirichlet problem given by Poisson's Integral Formula. If the boundary condition is given by $f(\zeta)$, then the solution to the Dirichlet problem is given by

$$u(z) = \begin{cases} \frac{1}{2\pi}\int_0^{2\pi} f(e^{it})\frac{1-|z|^2}{|1-ze^{-it}|^2}\,dt & \text{if } z \in D \\ f(z) & \text{if } z \in \partial D. \end{cases}$$

Question : When are two annuli conformally equivalent?
They are conformally equivalent if the ratio of their radii is the same. This is clearly a sufficient condition, for example consider the function $f(z) = \lambda z$ which takes $A(r, R)$ to $A(r, \lambda R)$. It turns out this condition is also necessary.

4 Analytic Number Theory

4.1 Circle Methods

Question : How many ways can you write a positive integer N as a sum of s kth powers?
By Hardy-Littlewood, we have

$$r(N) \sim C_{k,s}N^{\frac{s}{k}-1}\mathfrak{S},$$

as $N \to \infty$, where $\mathfrak{S}$ is the singular series.
Question : How large must s be for an asymptotic like this to hold?
Hardy and Littlewood proved one can take $s \geq (k-2)2^{k-1}+5$. Using Hua's Lemma, we can take $s \geq 2^k + 1$.
Question : Tell us about Weyl differencing. What bounds can you get?
For example, if I have a function $T(X) = \sum_{1\leq x\leq X} e(\psi(x))$, we can write

$$\begin{aligned} |T(X)|^2 &= \sum_{1\leq x,y\leq X} e(\psi(y)-\psi(x)) \\ &= \sum_{|h|<X} \sum_{\substack{1\leq x\leq X \\ 1\leq x+h\leq X}} e(\psi(x+h)-\psi(x)). \end{aligned}$$

Effectively Weyl differencing provides a method to decrease the complexity of the function ψ. In most cases $\psi(x)$ is a polynomial and Weyl differencing gives a way to only consider exponential sums of linear functions. The Weyl differencing lemma tells us in general that the bound we should expect is

$$|F(\psi)|^{2^J} \leq (2X)^{2^J-J-1} \sum_{|h_1|\leq X} \cdots \sum_{|h_j|\leq X} \sum_{x\in I_j(\vec{h})} e(\Delta_j(\psi(x);\vec{h})),$$

where the notation $\Delta_j(\psi(x),\vec{h})$ just means to apply Weyl differencing j times to ψ, which produces j parameters $h_1,\ldots,h_j$ which we write as the vector $\vec{h} = (h_1,\ldots,h_j)$.
Question : What's Hua's Lemma? Can you prove it?

Sure. So Hua's Lemma states that if $f(\alpha) = \sum_{1\le x\le X} e(\alpha x^k)$ then for $k \ge 2$ and $1 \le j \le k$ we have

$$\int_0^1 |f(\alpha)|^{2^J}\, d\alpha \ll X^{2^J - J + \epsilon}.$$

The proof, like Weyl's differencing lemma is by induction. First, we have

$$\int_0^1 |f(\alpha)|^2\, d\alpha = \int_0^1 f(\alpha)f(-\alpha)\, d\alpha = \#\{1 \le x, y \le X : x^k = y^k\} = \lfloor X \rfloor.$$

Now suppose the bound holds for all $1 \le j \le J$, with $J < k$. We want to show it's true for $J + 1$. We have

$$\begin{aligned}\int_0^1 |f(\alpha)|^{2^{J+1}}\, d\alpha &= \int_0^1 |f(\alpha)|^{2^J}|f(\alpha)|^{2^J}\, d\alpha = \int_0^1 f(\alpha)^{2^{J-1}} f(-\alpha)^{2^{J-1}} |f(\alpha)|^{2^J}\, d\alpha \\ &\le (2X)^{2^J - J - 1} \sum_{\substack{|h_i|\le X \\ x\in I_J(\vec{h})}} \int_0^1 f(\alpha)^{2^{J-1}} f(-\alpha)^{2^{J-1}} e(\alpha\Delta_J(x^k, \vec{h}))\, d\alpha,\end{aligned}$$

after applying the Weyl differencing lemma. By orthogonality this is bounded by the number of integral solutions to

$$\sum_{i=1}^{2^{J-1}} (u_i^k - v_i^k) = \Delta_j(x^k, \vec{h})$$

with $1 \le u_i, v_i \le X$. The first type of solutions are for

$$\sum_{i=1}^{2^{J-1}} (u_i^k - v_i^k) = 0,$$

but the number of solutions to this equation is given by the integral

$$\int_0^1 f(\alpha)^{2^{J-1}} f(-\alpha)^{2^{J-1}} = \int_0^1 |f(\alpha)|^{2^J}\, d\alpha \ll X^{2^J - J + \epsilon},$$

by the induction hypothesis. Meanwhile, $\Delta_J(x^k, \vec{h}) = 0$ means that

$$h_1 \cdots h_J \left(\frac{k!}{(k-J)!} x^{k-J} + \dots \right).$$

Then one of the h_i is zero or x satisfies a polynomail determined by $h_1, \dots, h_j$. Altogether there are $O(X^J)$ choices. So in total we have $X^J \cdot X^{2^J - J + \epsilon} = X^{2^J + \epsilon}$. Now we count the nonzero solutions. For a fixed choice of u_i, v_i we have

$$\sum_{i=1}^{2^{J-1}} (u_i^k - v_i^k) = N$$

and we'd like to count the number of solutions to

$$h_1 \cdots h_J \left(\frac{k!}{(k-J)!} x^{k-J} + \dots \right) = N.$$

Then the number of choices for $h_1, \dots, h_J$ and x is $O(\tau_{J+1}(N)) \ll X^\epsilon$. So the total contribution is

$$\sum_{\substack{1 \le u_i, v_i \le X \\ 1 \le i \le 2^{J-1}}} O(\tau_{J+1}(N)) \ll (X^2)^{2^{J-1}} X^\epsilon = X^{2^J + \epsilon}.$$

To finish the proof we simply plug this in to our estimate from the beginning to obtain

$$\int_0^1 |f(\alpha)|^{2^J} \, d\alpha \ll (2X)^{2^J - J - 1} X^{2^J + \epsilon} = X^{2^{J+1} - (J+1) + \epsilon}.$$

Question : Can you prove the bound you used for the divisor function?
Question : Talk about three primes. Why can't we use the same method to solve Goldbach?
We consider the function $F_N(x) = \sum_{p \le N} \log p e(px)$. We want to show that the integral

$$\int_0^1 F_N(x)^3 e(-Nx) \, dx,$$

tends to infinity as N gets large. This will show that every sufficiently large number N is the sum of three primes. The idea is that $|F_N(x)|$ is large when x is close to a rational number. For example, $F_N(1/2) = \sum_{p \le N} \log p e(p/2) = \log 2 - \sum_{3 \le p \le N} \log p$. So we separate the integral into major and minor arcs. The major arcs are actually a small density set of $[0,1]$ but are called major because of their contributions to the integral. The minor arcs on the other hand are "most" of [0,1], but the contribution from these arcs should be small. We define the major arcs to be

$$M_{a,q} = \{x \in [0,1] : |x - a/q| < Q/N\},$$

where $Q = \log^B N$, so that Q is in the range for using Siegel-Walfisz, $a \in [q]$ and $q \in [Q]$. So the major arcs $\mathfrak{m}$ are the union of each of the $M_{a,q}$'s. The contribution from the major arcs is an explicit constant coming from the singular series times N^2, with an error term of $o(N^2)$. So it would be good enough to show that the contribution from the minor arcs is $o(N^2)$. The strategy is to execute a balancing act between the L^∞ and L^2 norms of F. We have

$$\begin{aligned} \left| \int_{\mathfrak{m}} F_N(x)^3 e(-Nx) \, dx \right| &\le \max_{x \in \mathfrak{m}} |F_N(x)| \left(\int_{\mathfrak{m}} |F_N(x)|^2 \, dx \right) \\ &\le \max_{x \in \mathfrak{m}} |F_N(x)| \int_0^1 |F_N(x)|^2 \\ &\le \max_{x \in \mathfrak{m}} |F_N(x)| N \log N, \end{aligned}$$

where the last thing is by the prime number theorem and partial summation. Vinogradov showed that the L^∞ norm is $o(N/\log^D N)$ which is good enough to prove the theorem. If we tried this for two primes instead of three, the problem becomes that the expected contribution from the minor arcs is now N instead of N^2. For a naive bound, one might try just sticking absolute values in, to obtain

$$\left| \int_{\mathfrak{m}} F_N(x)^2 e(-Nx) \, dx \right| \le \int_0^1 |F_N(x)|^2 \, dx = N \log N,$$

which is BIGGER than the main term. So then we try to do the same trick with L^∞ but now we need a good L^1 bound rather than a good L^2 bound. Even if we assume the average size of $|F_N(x)|$ is like $\sqrt{N}$, based on the L^2 norm, we would need $\|F_N(x)\|_\infty = o\left(\sqrt{\frac{N}{\log N}}\right)$ on the minor arcs. However, this is better than square root cancellation, and is not to be reasonably expected. Cauchy-Schwarz is also an idea, but you run into the same problem.

4.2 Classical Stuff

Question : Can you show the zeta function doesn't vanish on the line $\mathrm{Re}(s) = 1$**?**
Suppose $\zeta(1+it_0) = 0$ for some $t_0 \neq 0$. Then

$$|\zeta(\sigma + it_0)|^4 \leq C(\sigma - 1)^4,$$

as $\sigma \to 1$. Since ζ has a simple pole at $s = 1$, we also have

$$|\zeta(\sigma)|^3 \leq C'(\sigma - 1)^{-3},$$

as $\sigma \to 1$. Meanwhile, ζ is holomorphic at $\sigma + 2it_0$, so it stays bounded there. Altogether this implies that

$$|\zeta^3(\sigma)\zeta^4(\sigma+it_0)\zeta(\sigma+2it_0)| \to 0 \text{ as } \sigma \to 1.$$

However, the 3-4-1 inequality implies that

$$\log|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 0,$$

which is a contradiction since log is negative for values between 0 and 1.
Question : Can you give a zero-free region for the zeta function?
If $\zeta(\sigma+it) = 0$, then there is a constant $c > 0$ such that if $|t| \geq 2$,

$$\sigma < 1 - \frac{c}{\log|t|}.$$

By the 3-4-1 inequality, we have

$$3\left[-\frac{\zeta'}{\zeta}(\sigma)\right] + 4\,\mathrm{Re}\left[-\frac{\zeta'}{\zeta}(\sigma+it)\right] + \mathrm{Re}\left[-\frac{\zeta'}{\zeta}(\sigma+2it)\right] \geq 0.$$

In general, we have

$$-\frac{\zeta'}{\zeta}(s) = \frac{1}{s-1} + B_1 + \frac{1}{2}\frac{\Gamma'}{\Gamma}(s/2+1) - \sum_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho}\right).$$

Notice that

$$\mathrm{Re}\left(\frac{1}{\sigma+2it-\rho}\right) = \frac{\sigma-\beta}{(\sigma-\beta)^2 + (2t-\gamma)^2} \geq 0,$$

so we can throw away these terms. (Note we are writing $\rho = \beta + i\gamma$). Anyways, we have $\mathrm{Re}\left(-\frac{\zeta'}{\zeta}(\sigma+2it)\right) < O(\log|t|)$. We also know that $\frac{\zeta'}{\zeta}(\sigma) = \frac{1}{\sigma-1} + O(1)$. Finally, let $\rho = 1 - \delta + it$. If we throw out all the terms in the sum over zeros, except this one, we obtain

$$-\mathrm{Re}\,\frac{\zeta'}{\zeta}(\sigma+it) < O(\log|t|) - \frac{1}{\sigma - 1 + \delta}.$$

The 3-4-1 inequality thus gives that

$$\frac{4}{\sigma+\delta-1} < \frac{3}{\sigma-1} + O(\log|t|) \implies \delta \gg \frac{1}{\log t}.$$

Question : Define a Dirichlet L-function
The Dirichlet L-function associated to a Dirichlet character χ is

$$L(s,\chi) = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1} = \sum_{n=1}^{\infty} \chi(n) n^{-s}.$$

It converges for $\mathrm{Re}(s) > 1$.
Question : Can you extend the convergence of a Dirichlet L-function?
WHen χ is a non-principal Dirichlet character then $L(s,\chi)$ extends to a holomorphic function on $\mathrm{Re}(s) > 0$. Suppose χ is a Dirichlet character modulo q. Let $A(x) = \sum_{n\leq X} \chi(n)$. Then $A(x+q) - A(x) = \sum_{n\in(\mathbb{Z}/q\mathbb{Z})} \chi(n) = 0$, so $A(x)$ is periodic and bounded. By partial summation, we have

$$\begin{aligned} L(s,\chi) = \sum_{n\geq 1} \chi(n) n^{-s} &= x^{-s} A(x) \mid_0^\infty - \int_0^\infty A(t)(-st^{-s-1})\, dt \\ &= s \int_0^\infty A(t) t^{-s-1}\, dt, \end{aligned}$$

which extends to a holomorphic function on $\mathrm{Re}(s) > 0$ since it is the limit of the uniformly converging sequence of functions $\phi_n(s) = s\int_0^n A(x) x^{-s-1}\, dx$ since $A(x)$ is bounded.
Question : Why are there infinitely many primes congruent to $1 \bmod q$?
The key is to show that $L(1,\chi) \neq 0$. We motivate it with the 1 mod 4 case. We have

$$\log L(s,\chi) = \sum_{p\equiv 1 \bmod 4} p^{-s} - \sum_{p\equiv 3 \bmod 4} p^{-s} + O(1),$$

and also

$$\log\zeta(s) = \sum_p p^{-s} + O(1),$$

so

$$\frac{\log\zeta(s) + \log L(s,\chi)}{2} = \sum_{p\equiv 1 \bmod 4} p^{-s} + O(1).$$

The idea is to show the RHS diverges. We already know $\log\zeta(s) \to \infty$ as $s \to 1^+$, so it would suffice to have that $\log L(s,\chi) = O(1)$. In particular, $L(s,\chi) \neq 0$ since otherwise the $\log L(s,\chi)$ term would tend to negative infinity. The trick is that the Dedekind zeta function of a number field K factors as a zeta function and the L-function. Since $\zeta(s)$ has a pole at $s = 1$, if one could show that $\zeta_K(s)$ extends to a meromorphic function witha simple pole at $s = 1$, then we could conclude that $L(s,\chi)$ extends to a meromorphic function containing $s = 1$ and moreover that $\mathrm{ord}_{s=1} L(s,\chi) = 0$. One can show directly that $\zeta_K(s)$ extends to a meromorphic function on $\mathrm{Re}(s) > 1/2$ with a simple pole at $s = 1$, but this more interestingly follows from the analytic class number formula which shows that $\lim_{s\to 1^+}(s-1)\zeta_K(s)$ is a nonzero constant depending on a number of invariants of the number field K.

4.3 Sieve Methods

5 Representation Theory of Compact Lie Groups

Question : Define maximal tori.
A maximal torus is a compact, connected abelian subgroup that is a maximal such subgroup with respect to inclusion.
Question : Ok then why are they called tori?
If T is a torus with $\dim T = k$ then $T \cong \mathbb{R}^k/\mathbb{Z}^k$. Indeed, let $\mathfrak{t} = \mathrm{Lie}(T)$ be the lie algebra and note that the exponential map $\exp\colon \mathfrak{t} \to T$ is surjective. Then $T \cong \mathfrak{t}/\ker(\exp)$. Since $\ker(\exp)$ is discrete, $\ker(\exp) \cong \mathbb{Z}^r$ for some $r \leq k$, a priori. And also note that $\mathfrak{t} \cong \mathbb{R}^k$, so $T \cong (\mathbb{R}/\mathbb{Z})^r + \mathbb{R}^{k-r}$, but since T is supposed to be compact, we must have $r = k$, so that $T \cong (\mathbb{R}/\mathbb{Z})^k$.

5.1 Lie groups and Lie algebras

Question : How do you get from a Lie algebra to a Lie group?
If G is a closed Lie subgroup of $\mathrm{GL}_n(\mathbb{C})$, then let Lie(G) be the set of all $X \in M_{n\times n}(\mathbb{C})$ such that $\exp(tX) \subset G$. Then Lie(G) is a Lie subalgebra of $\mathfrak{gl}_n(\mathbb{C})$ and we obtain the Lie group by exponentiating this Lie algebra.
Question : What is the exponential map more generally?
Let G be a Lie group and let $\mathfrak{g} = T_1G$, the tangent space at the identity. Then for $x \in \mathfrak{g}$, there is a unique one-parameter subgroup $\gamma_x\colon \mathbb{R} \to G$ such that $\gamma_x'(0) = x$. Then the exponential map $\exp\colon \mathfrak{g} \to G$ is defined by

$$\exp(x) = \gamma_x(1),$$

where γ_x is the one-parameter subgroup with tangent vector at the identity equal to x.
Question : What is the exponential map from $\mathbb{R} \to \mathbb{R}$.?
Well the one parameter subgroup for $a \in \mathbb{R}$ is given by $\gamma_a(t) = ta$, so $\exp(a) = \gamma_a(1) = a$, so the exponential map is the identity.

5.2 Representations

Question : What is the Weyl character formula?
The Weyl character formula describes the characters of irreducible representations of compact Lie groups in terms of their highest weights. In particular, if Γ_λ is the irreducible representation with highest weight , then the character of this representation is given by the following formula :

$$\mathrm{char}(\Gamma_\lambda) = \frac{\sum_{w\in W}(-1)^w e((\lambda+\rho)w))}{\sum_{w\in W}(-1)^w e(w\rho)}.$$

In particular, to evaluate on an element, we only need to consider the torus. Let H be in the Lie algebra of the torus. Then the Weyl character formula tells us that

$$\chi(e^H) = \frac{\sum_{w\in W}(-1)^w e(\langle w(\lambda+\rho), H\rangle)}{\sum_{w\in W}(-1)^w e(\langle \rho, H\rangle)}.$$

The character formula is often written in the more succinct form $\mathrm{char}(\Gamma_\lambda) = \frac{A_{\lambda+\rho}}{A_\rho}$.
Question : Can you give an expression for the dimension of the representation

Γ_λ.
So the dimension of the irreducible representation Γ_λ with highest weight λ is given by :

$$\dim \Gamma_\lambda = \prod_{\alpha \in R^+} \frac{\langle \lambda + \rho, \alpha \rangle}{\langle \rho, \alpha \rangle},$$

where I will usually take $\langle , \rangle$ to be the standard inner product, but I suppose it is properly expressedf as $\langle \alpha, \beta \rangle = 2(\alpha, \beta)/(\beta, \beta)$ where $(,)$ is the Killing form.
Question : You've mentioned ρ twice, but haven't defined it. What is it?
It is the Weyl element or Weyl vector. I like to define it as the half sum of the positive roots, i.e.

$$\rho = \frac{1}{2} \sum_{\alpha \in R^+} \alpha = \frac{1}{2} \sum_{i<j} x_i - x_j,$$

in most cases.
Question : Why can we assume finite dimensional reps of a compact Lie group G are unitary?
Let (π, V) be a representation. Let $(,)$ be an arbitrary inner product on V. We can define a new inner product by

$$\langle v, w \rangle = \int_G (\pi(g)v, \pi(g)w)\, dg.$$

By the G-invariance of the Haar measure, this new inner product is unitary uder the action of $\pi(g)$.
Question : Give a representation of a compact lie group G
Consider the space $L^2(G)$ space of measurable functions on G such that $\int_G |f(g)|^2\, dg < \infty$. Consider the left regular representation $(L(g)f)(x) = f(g^{-1}x)$. The map $g \mapsto L(g)f$ is a continuous map from G to $L^2(G)$ and $g \mapsto L(g)$ is a representation $G \to U(L^2(G))$, since this operator is unitary. The big idea is that we can generate lots of representations of G by looking at the eigenspaces of an invariant convolution operator. Indeed, by restricting the regular representation to $U(V_\lambda)$ we will obtain a finite-dimensional representation of G. I think the left regular representation will work, but in general you can take T to be a Hilbert-Schmidt operator with G-invariant kernel K. T is nonzero, so by the Spectral theorem it has a nonzero real eigenvalue λ and the eigenspace V_λ is finite dimensional.
Question : How do you know the irreducible representations are finite dimensional?
We summarize an argument of Nachbin. We're given an irreducible unitary rep. Let w be a unit vector in V. Consider the map

$$T \colon v \mapsto \int_G gw \langle gw, v \rangle dg.$$

Then T is a G-invariant linear map so by Schur's lemma it is just scaling by some λ. To see λ is not zero, note that

$$\lambda = \langle w, \lambda w \rangle = \langle w, Tw \rangle = \int_G |\langle gw, w \rangle|^2\, dg > 0,$$

since the integral is positive in a neighborhood of the identity. So $\lambda \neq 0$ and so

$$\frac{1}{\lambda} \int_G gw \langle gw, \cdot \rangle dg,$$

is the identity. In fact this operator is trace class and so taking traces of both sides we see that $\dim V$ is finite.

Question : What is the Peter Weyl theorem?

First is says that the linear span of all matrix coefficients for all finite-dimensional irreducible unitary representations of G is dense in $L^2(G)$. Every unitary irreducible representation of G is finite dimensional, as we showed above. Lastly it says that the regular representation on $L^2(G)$ decomposes into a direct sum of irreducible unitary representations. IN fact, any unitary representation of G on a Hilbert space V will decompose into an orthogonal sum of finite-dimensional irreducible invariant subspaces (where each subspace V_λ appears $\dim V_\lambda$ many times.

Question : What is Weyl's unitarization trick?

It says that holomorphic representations of $\mathrm{SL}_2(\mathbb{C})$ are in correspondence with representations of $SU(2)$.

Question : Does $\mathrm{SL}_2(\mathbb{R})$ have any finite dimensional unitary reps?

No. Consider $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ which are all conjugate. Then there is some A_n such that

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = A_n \begin{pmatrix} 1 & 1/n \\ 0 & 1. \end{pmatrix} A_n^{-1}$$

Let ρ be a unitary representation. Then taking limits as $n \to \infty$ and taking ρ of both sides shows that ρ is the identity on this conjugacy class. Then you can do the same thing for lower triangular matrices.